COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	08/26/26	**DATE SUBMITTED:**	08/26/26
NAME:	Villaruel, Prince Benjamin S.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. **Prelim Laboratory Exam Set D**

1. PROCEDURE OUTPUT

What procedures did you perform?

Algorithm

A. Stack using Array

1. Start
2. Set a max cap.
3. Input test cases.
4. Check if the sequence is symmetric using a stack.
5. Compare the original sequence.
6. Display if the following sequence is valid or not.
7. End

Analysis: For stacks, I used array-based implementation since I already have a code snippet that I can easily modify to validate the following sequence. For the first step, since it is array-based, I first set a max cap for the stack array. Second, I added the following test cases in the main function. For checking and comparing, I used for loops to browse each test case, followed by if statements to determine if it is symmetric by focusing checking the top . Finally, after each function validates if the following test cases are symmetric, I display the results in the main function.

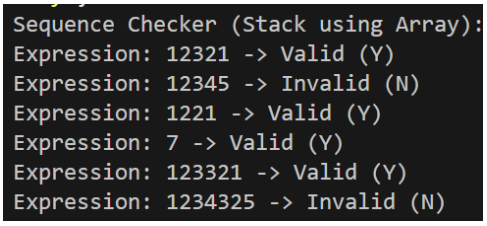
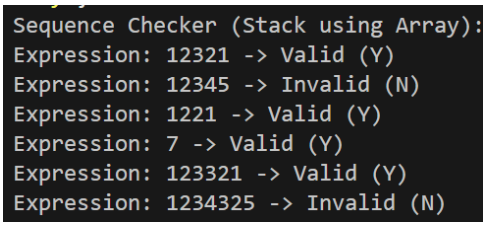
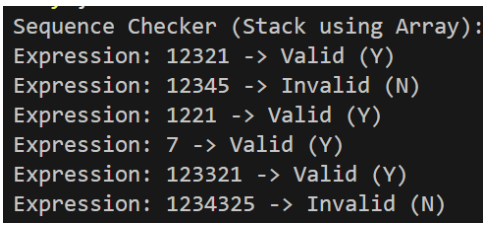
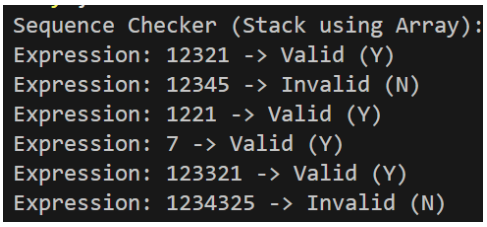
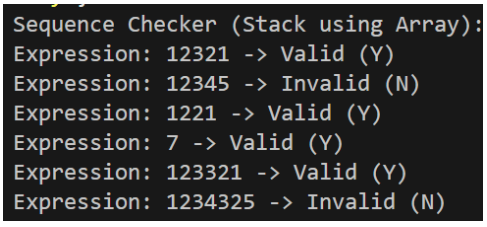
B. Queue using Linked List

1. Start
2. Create a header file for
3.
4. Create the main function for the output.
5. End

Analysis:

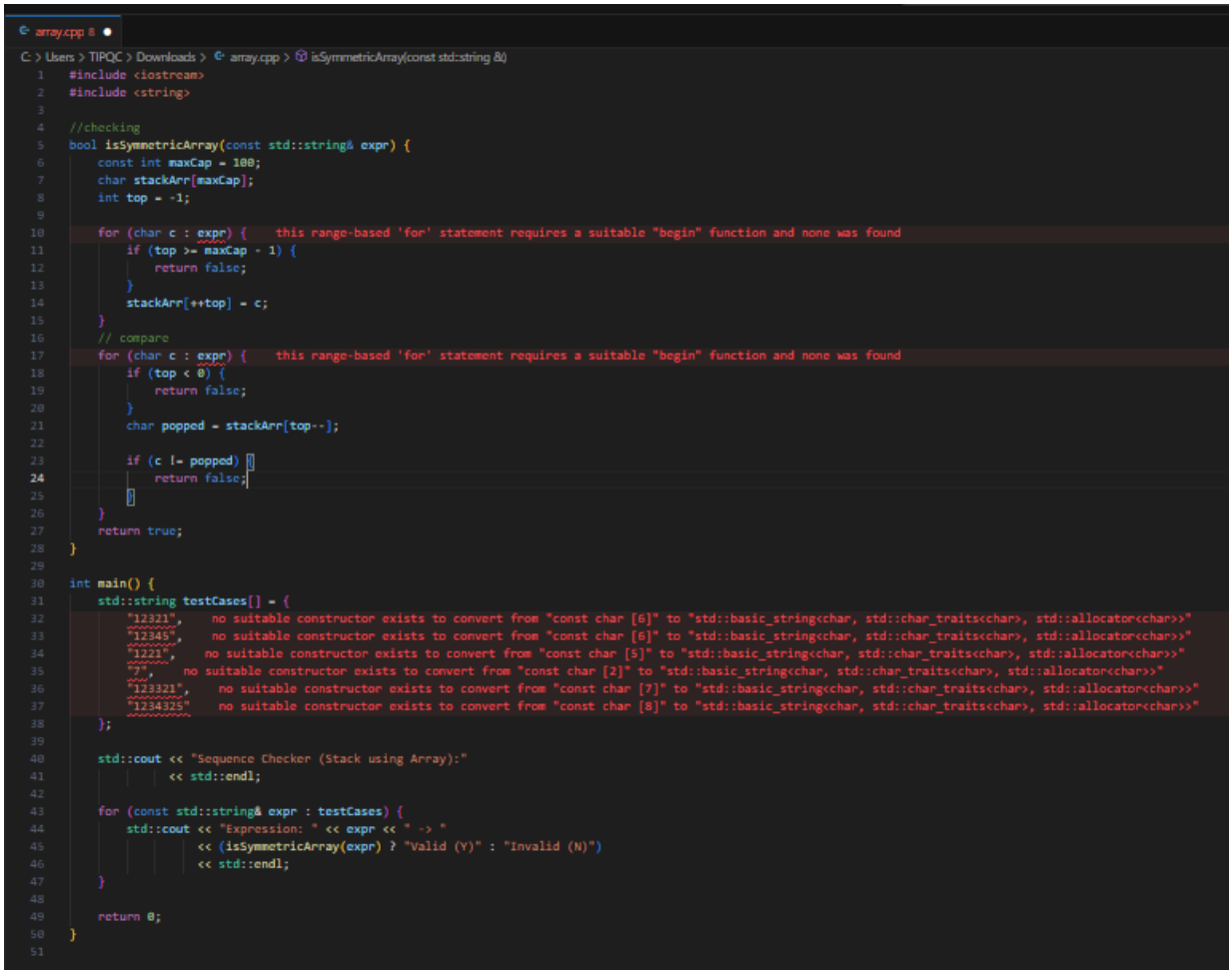
Sample Test Data (Stack using Array)

Input Sequence	Symmetric?	Output (Console)	Remark

12321	Yes		The stack (array-based implementation) successfully displayed the same results as stated in the sample test data which shows that it is symmetric as it meets the conditions.
12345	No		The stack (array-based implementation) successfully displayed the same results as stated in the sample test data which shows that it is not symmetric as it meets the conditions.
1221	Yes		The stack (array-based implementation) successfully displayed the same results as stated in the sample test data which shows that it is symmetric as it meets the conditions.
7	Yes		The stack (array-based implementation) successfully displayed the same results as stated in the sample test data which shows that it is symmetric as it meets the conditions.
123321	Yes		The stack (array-based implementation) successfully displayed the same results as stated in the sample test data which shows that it is symmetric as it meets the conditions.
1234325	No		The stack (array-based implementation) successfully displayed

		Sequence Checker (Stack using Array): Expression: 12321 -> Valid (Y) Expression: 12345 -> Invalid (N) Expression: 1221 -> Valid (Y) Expression: 7 -> Valid (Y) Expression: 123321 -> Valid (Y) Expression: 1234325 -> Invalid (N)	the same results as stated in the sample test data which shows that it is not symmetric as it meets the conditions.

Code Snippet



```cpp
#include <iostream>
#include <string>

//checking
bool isSymmetricArray(const std::string& expr) {
    const int maxCap = 100;
    char stackArr[maxCap];
    int top = -1;

    for (char c : expr) {    this range-based 'for' statement requires a suitable "begin" function and none was found
        if (top >= maxCap - 1) {
            return false;
        }
        stackArr[++top] = c;
    }
    // compare
    for (char c : expr) {    this range-based 'for' statement requires a suitable "begin" function and none was found
        if (top < 0) {
            return false;
        }
        char popped = stackArr[top--];

        if (c != popped) {
            return false;
        }
    }
    return true;
}

int main() {
    std::string testCases[] = {
        "12321",    no suitable constructor exists to convert from "const char [6]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
        "12345",    no suitable constructor exists to convert from "const char [6]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
        "1221",    no suitable constructor exists to convert from "const char [5]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
        "7",    no suitable constructor exists to convert from "const char [2]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
        "123321",    no suitable constructor exists to convert from "const char [7]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
        "1234325"    no suitable constructor exists to convert from "const char [8]" to "std::basic_string<char, std::char_traits<char>, std::allocator<char>>"
    };

    std::cout << "Sequence Checker (Stack using Array):"
              << std::endl;

    for (const std::string& expr : testCases) {
        std::cout << "Expression: " << expr << " -> "
                  << (isSymmetricArray(expr) ? "Valid (Y)" : "Invalid (N)")
                  << std::endl;
    }

    return 0;
}
```

Sample Test Data (Queue using Linked List)

Input Sequence	Symmetric?	Output (Console)	Remark

12321	Yes		
12345	No		
1221	Yes		
7	Yes		
123321	Yes		
1234325	No		

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.